

AI Content Performance Analysis Using Python and Power BI

1. Introduction

Artificial Intelligence (AI) is widely used for creating digital content such as text, images, and videos. This project analyzes the performance of AI-generated content using Python and Power BI. The aim is to understand content performance, compare platforms, and provide business insights using data analysis and visualization.

2. Objectives

- To analyze AI content performance.
- To clean and preprocess the dataset using Python.
- To handle missing values.
- To perform Exploratory Data Analysis (EDA).
- To build an interactive Power BI dashboard.
- To generate business insights and recommendations.

3. Dataset Description

The dataset contains information about AI-generated content published on different platforms.

Important columns include:

- Content_ID.
- Platform
- AI_Tool_Used
- AI_Type
- Content_Type
- Duration_Seconds
- Total_Views
- Total_Likes
- Total_Shares
- Total_Comments
- Engagement_Rate
- Production_Cost_USD
- Revenue_USD
- Watch_Time_Avg
- Retention_Rate

4. Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Excel
- Power BI

5. Methodology

The project was completed in the following steps:

- Loaded the dataset into Python.
- Cleaned the data.
- Handled missing values.
- Checked data types.
- Performed Exploratory Data Analysis (EDA).
- Created Python visualizations.
- Exported the cleaned dataset to Excel.
- Imported the cleaned dataset into Power BI.
- Built an interactive dashboard.
- Generated business insights and recommendations.

6. Data Cleaning

The dataset was cleaned using Python.

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked missing values.
- Imputed missing values.
- Verified data types.
- Removed data inconsistencies.
- Prepared the dataset for visualization.

7. Python Visualizations

The following charts were created using Python:

understand the dataset before building the Power BI dashboard.

Brief Explanation

- **Count Plot** – Displays the count of each category in the dataset.
- **Histogram Plot** – Shows the distribution of numerical values.
- **Box Plot** – Identifies outliers and shows data distribution.
- **Violin Plot** – Displays both the distribution and density of data.
- **Bar Chart** – Compares values across different categories.
- **Horizontal Bar Chart** – Compares categories using horizontal bars.
- **Pie Chart** – Shows the percentage share of different categories.
- **Scatter Plot** – Shows the relationship between two numerical variables.
- **Correlation Heatmap** – Displays the correlation between numerical variables.
- **Area Chart** – Shows trends and cumulative values over data.
- **Pivot Table** – Summarizes data for comparison and analysis.
- **Pivot Table Heatmap** – Highlights important values in the pivot table using color intensity.

8. Power BI Dashboard

Two dashboards were created.

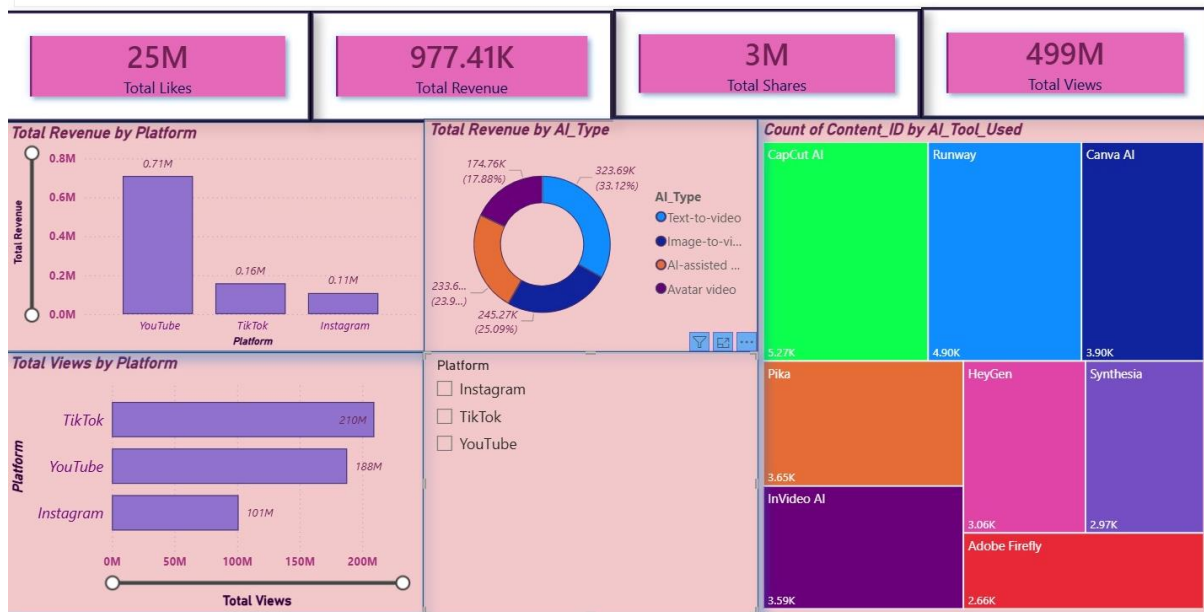
Dashboard 1 – AI Content Performance Dashboard

This dashboard includes:

- Total Views
- Total Likes
- Total Shares
- Total Revenue
- Total Views by Platform
- Total Revenue by Platform
- Revenue by AI Type
- Content Count by AI Tool

This dashboard provides an overview of AI content performance.

AI Content Performance Dashboard

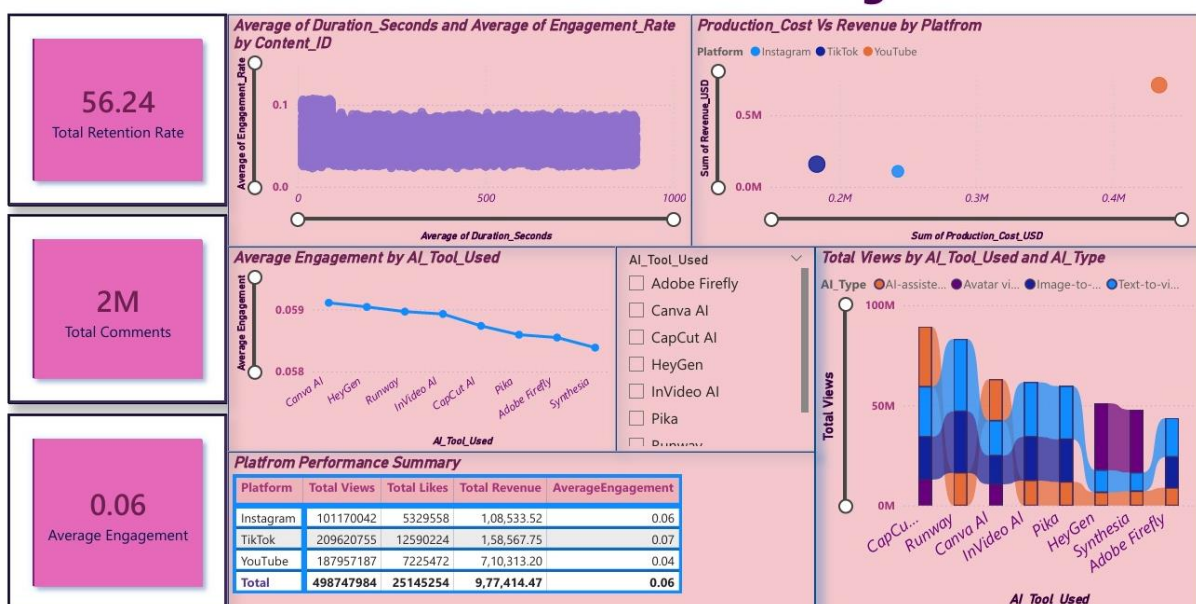


Dashboard 2 – AI Content Performance Insights

This dashboard includes:

- Duration vs Engagement Rate
- Production Cost vs Revenue
- Average Engagement by AI Tool
- Platform Performance Matrix
- This dashboard provides detailed business insights.

AI Content Performance Insights



9. Business Insights

1. High Audience Reach

TikTok recorded the highest total views (209.62M), indicating that it is the most effective platform for reaching a larger audience and increasing brand visibility.

2. Highest Revenue Platform

YouTube generated the highest total revenue (710.31K USD), showing that it is the most profitable platform for AI-generated content and should be prioritized for revenue-focused campaigns.

3. Most Popular AI Tool

CapCut AI was the most frequently used AI tool (5.27K contents), indicating its popularity and effectiveness in creating AI-generated content efficiently.

4. Best Performing AI Content Type

Text-to-Video AI generated the highest revenue (323.69K USD), suggesting that video-based AI content delivers better financial performance than other AI content types.

5. Platform Performance Difference

The analysis shows that different platforms perform differently. TikTok leads in views and engagement, while YouTube leads in revenue. This highlights the importance of selecting the right platform based on business objectives.

6. Interactive Dashboard

The Power BI dashboard provides a clear comparison of platform performance, AI tools, and AI content types, helping businesses make faster and more effective data-driven decisions

10. Final Recommendations

Recommendation 1

TikTok recorded the highest total views (209.62M) and the highest average engagement rate (0.07). Therefore, businesses should prioritize TikTok for campaigns focused on maximizing audience reach and audience engagement.

Recommendation 2

YouTube generated the highest total revenue (710.31K USD). Organizations aiming to maximize business revenue should focus on creating high-quality video content and monetization strategies on YouTube.

Recommendation 3

CapCut AI was the most frequently used AI tool (5.27K contents), indicating its effectiveness and popularity in AI content creation. Businesses should prioritize CapCut AI to improve content creation efficiency and maintain consistent content quality.

Recommendation 4

The Text-to-Video AI content type generated the highest revenue (323.69K USD). Companies should invest more in Text-to-Video content because it has demonstrated the strongest financial performance among all AI content types.

Recommendation 5

Based on the dashboard analysis, businesses should adopt a platform-specific strategy by using TikTok for increasing audience reach and engagement, YouTube for maximizing revenue, and improving Instagram content strategy to enhance its overall performance.

11. Conclusion

- This project analyzed the performance of AI-generated content across different social media platforms using Python and Power BI.
- The analysis revealed that TikTok achieved the highest audience reach and engagement, while YouTube generated the highest revenue, making it the most profitable platform.
- Among AI tools, CapCut AI was the most frequently used for content creation, and Text-to-Video AI produced the highest revenue among all AI content types.
- These findings show that selecting the right platform, AI tool, and content type plays a significant role in improving audience engagement, increasing revenue, and enhancing overall content performance.
- The insights and recommendations from this analysis can help businesses and content creators make data-driven decisions to optimize their AI content strategies.